CATERPILLAR®

BEST PRACTICE SERIES

Best Practice Electrical System Preventive Maintenance

Application | **Maintenance** | Site Management | Component Rebuild | Component Life Management | MARC | Management

CAT® Global Mining

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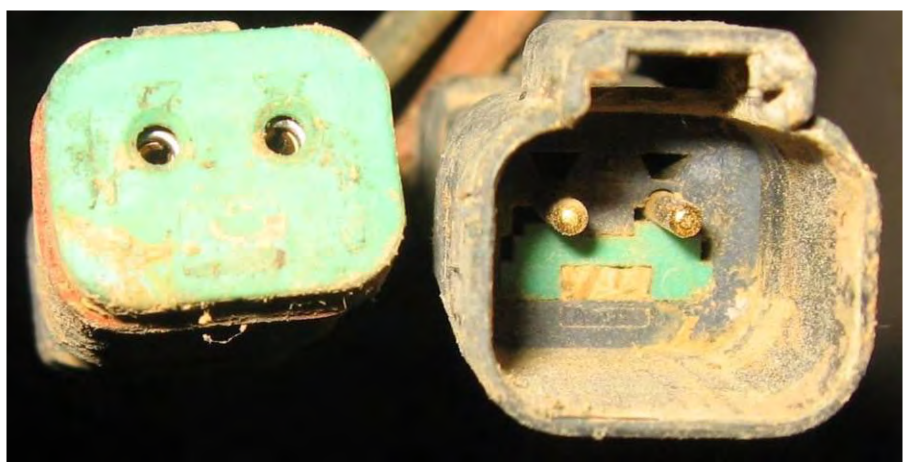
1.0 Introduction

This Best Practice provides an overview of one strategy for managing electrical systems on Caterpillar mining equipment operating in an environment with extreme conditions that impact the reliability of electrical systems.

The previously mentioned extreme conditions consist of factors which may permit water to enter into electrical systems, mainly at electrical connectors or connections. These factors may include extreme rainfall or humidity, substandard maintenance practices or machine wash processes that use high pressure, high pH water. These factors can result in water entering into electrical connectors and components which causes both immediate and future issues. Immediate problems are due to the presence of water resulting in short circuit faults. Future problems are due to the presence of corrosive residues resulting in open circuit faults. In both cases the basic cause is the same being that water has been allowed to enter into the electrical connectors. In both failure modes the result is unscheduled equipment downtime and production loss. These events require maintenance intervention which if not executed to an acceptable standard can impact future diagnostics, increase operating costs and impact the reliability and availability of equipment.

2.0 Best Practice Description

Generally electrical systems are managed through corrective maintenance and occasionally through periodic maintenance strategies. Corrective maintenance activities, random failure rate patterns, and substandard maintenance practices can negatively effect electrical system performance and can cause electrical components to not achieve their expected life (PCR). The implementation of preventative maintenance practices can improve the electrical system reliability and improves the probability that components will achieve or exceed their expected life or PCR. This preventative maintenance process includes various tasks to provide protection to electrical components.



3.0 Implementation Steps

This Continuous Improvement initiative began with the identification of repetitive electrical problems and high cost per hour related to electrical systems. The proposed solution was presented based on positive experiences from other mines and applications.

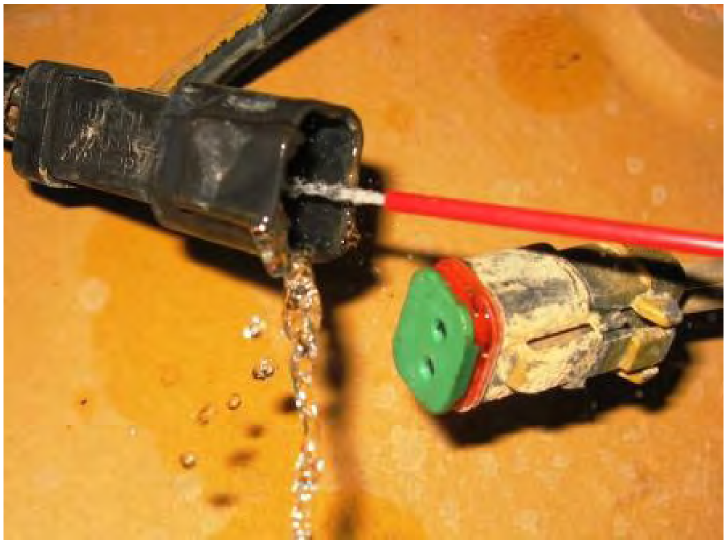
This program was not managed as a 6 Sigma project because the solution was known and validated in other organizations. The implementation took approximately six months and was lead by Reliability Engineering. Change was managed through communication and socialization at all operational levels with positive results and “buy-in”.
All organizational culture and process changes provide challenges related to acceptance and must be managed through a collaborative and constant communication.

In regards to electrical connectors the process begins with the cleaning or removal of contamination or corrosion from the connectors. The second step involves applying specialized grease which contains Vapor Phase Corrosion Inhibitors (VPCI) to the connector. This grease provides two benefits. The first benefit is a physical barrier preventing any water that enters the connector from making contact with the metal pins & sockets. The second benefit of the grease is its VPCI action which prevents oxidation of the metallic parts if water does make contact with the metallic parts. The final step of the process for protecting electrical connectors involves covering the connector with a self vulcanizing tape which provides an external physical barrier to prevent water from entering the connectors.

Senson Kit

Free the Connector

Clean the Connector



Use contact cleaner to eliminate contamination & corrosion.

Allow the solvent evaporate.

Apply the Senson Grease

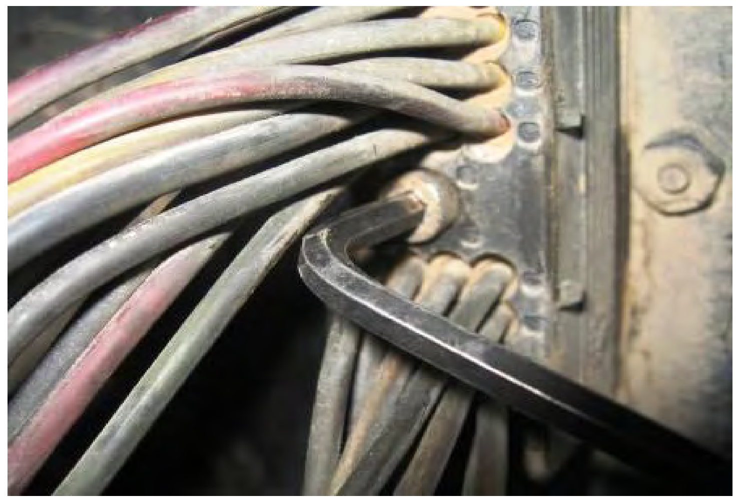
Use sufficient grease to cover the connector pins.

Avoid hydraulic lock. Too much grease could crack the connector.

Internal Protection




Re-connect



Ensure that each connector is correctly coupled

Cut Vulcanizing Tape

Approximately 200mm

External Protection

Clean the outside of the connector before applying the vulcanizing tape.

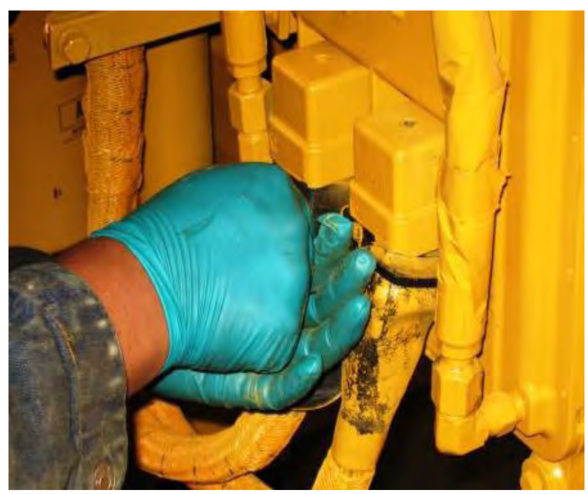
External Protection

Always cover the connector with vulcanizing tape where practical

Secure Connector

Always secure connectors in their bracket or with a zip tie to avoid harness chaffing or other physical damage.

793D Engine: ECM Protection

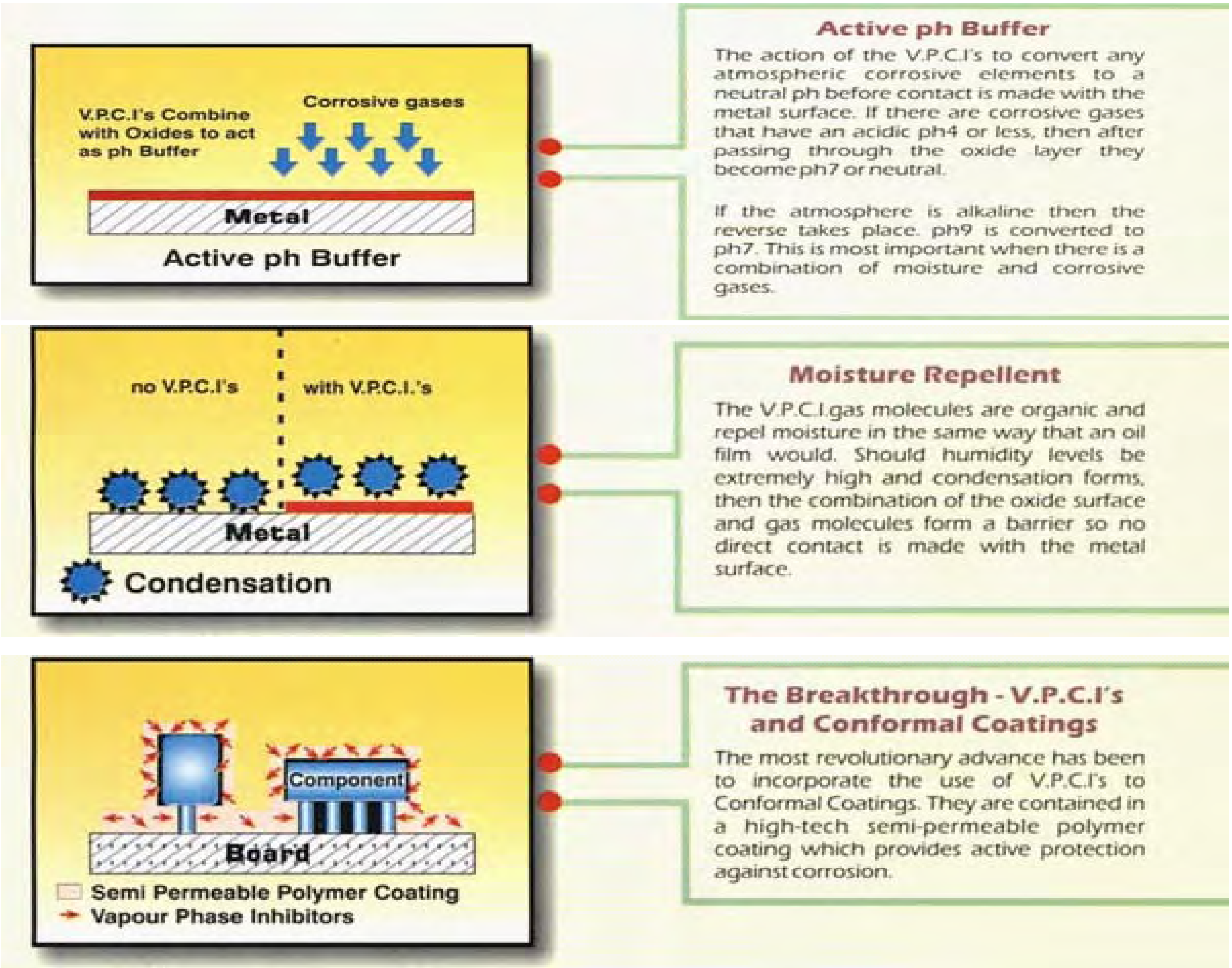


Apply Electroguard Grease in connectors J1 & J2 . Secure rubber boots with zip ties

<u>Battery Protection</u>

Clean Battery post and clamp with a wire brush, tighten the clamp bolt, cover the battery terminal with Electroguard Grease.

A third part of the process consists of protecting electrical components in closed compartments, such as electronic control modules. Humidity in closed compartments can cause oxidation of exposed surfaces. In these applications Vapaguard pads or capsules are installed according to the cubic capacity of the compartment. These products also utilize VPCI technology which emits a non-toxic anti-corrosive vapor preventing the oxidation of exposed surfaces. These products have an installed life of twelve months.

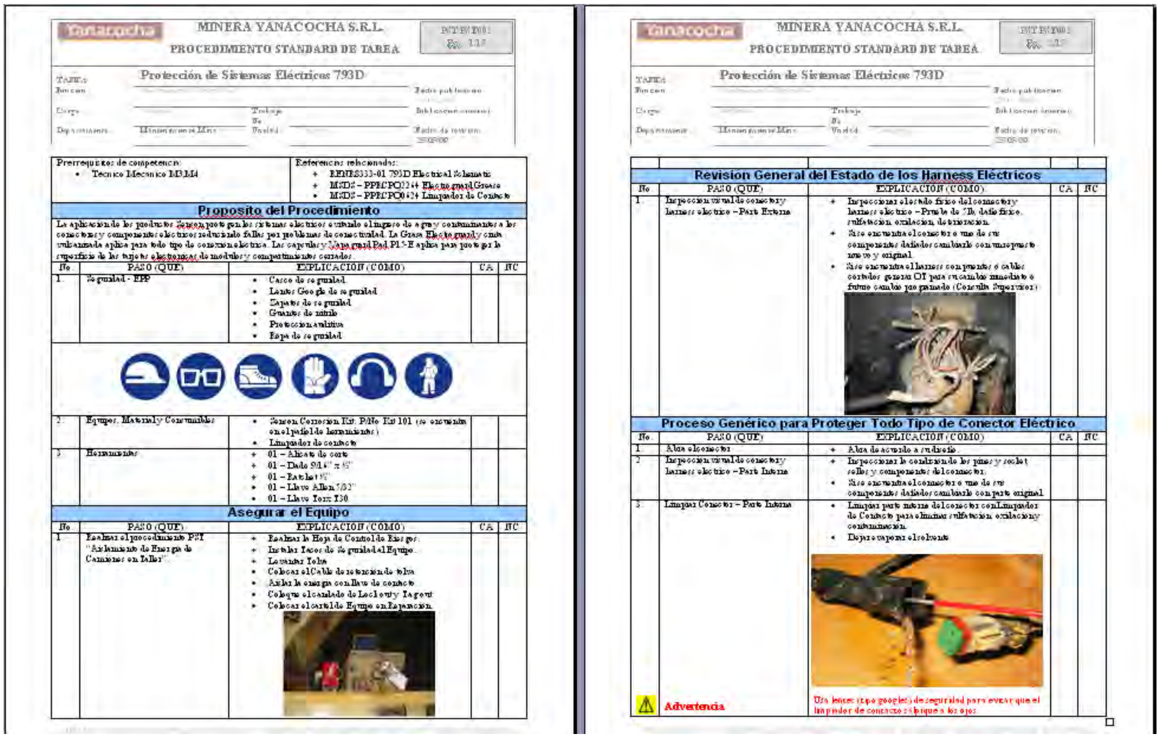


Corrosion Protection for Electronic & Electrical Units

Three Vapaguard pads installed in the compartment at the rear of a 793C Cab.

One Vapaguard pad installed in the gear-shift console in a 793C Cab

One Vapaguard capsule installed in the closed compartment in the roof of a 793C Cab


MINERA YANACOCHA S.R.L.
PROCEDIMIENTO STANDARD DE TAREA
Protección de Sistemas Eléctricos 793D
Proposito del Procedimiento
Asegurar el Equipo
Revision General del Estado de los Harness Eléctricos
Proceso Genérico para Proteger Todo Tipo de Conector Eléctrico
Advertencia

Standard procedures are attached.

4.0 Benefits

The perceived benefit of this process is the elimination of electrical failures due to water entering the system via the connectors. This will result in less unscheduled downtime due to electrical failures which will in turn lower overall O&O costs.

5.0 Resources Required

The estimated investment associated to this project is approximately US$200 per OHT in materials and 48 man hrs. per year of preventative maintenance which can be planned and executed in parallel with other major tasks to avoid additional downtime.

For specific information on the kit used in this Best Practice please refer to the files attached. Materials similar to those used in this kit may be available for purchase locally.

6.0 Supporting Attachments / References

- None

7.0 Related Best Practices

- None

8.0 Acknowledgements

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